Trigonometric Equation: Questions and Full Solutions

Question 01

English

Let λ be a real number with $\lambda < -2$. Then the sum of all the principal solutions (that is, the solutions lying in $[0, 2\pi)$) of the equation $\tan x + \cot x = \lambda$ is equal to

(A) 3π
(B) 4π
(C) 5π
(D) 6π

Only one option is correct.

हिन्दी

माना कि λ एक वास्तविक संख्या है जिसके लिए $\lambda < -2$ है। तब समीकरण $\tan x + \cot x = \lambda$ के सभी मुख्य हलों (अर्थात् $[0, 2\pi)$ में स्थित हलों) का योग बराबर है

(A) 3π
(B) 4π
(C) 5π
(D) 6π

केवल एक विकल्प सही है।

Solution

For x not a multiple of $\frac{\pi}{2}$: $\tan x + \cot x = \dfrac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \dfrac{2}{\sin 2x}$.

So the equation is $\sin 2x = \dfrac{2}{\lambda}$. Since $\lambda < -2$, $\dfrac{2}{\lambda} \in (-1, 0)$; write $\sin 2x = -k$ with $0 < k < 1$.

Let $\theta = \sin^{-1} k$, so $0 < \theta < \frac{\pi}{2}$. The general solution of $\sin 2x = \sin(-\theta)$ is $2x = n\pi + (-1)^n(-\theta)$, $n \in \mathbb{Z}$.

A principal solution satisfies $0 \le x < 2\pi$, i.e. $0 \le 2x < 4\pi$. In $[0, 4\pi)$ the sine is negative only on $(\pi, 2\pi)$ and $(3\pi, 4\pi)$, and on each of these it takes the value $-k$ twice:

$2x = \pi + \theta,\ 2\pi - \theta$ (from $n = 1, 2$) and $2x = 3\pi + \theta,\ 4\pi - \theta$ (from $n = 3, 4$).

None of these x is a multiple of $\frac{\pi}{2}$ (that would need $\sin 2x = 0$), so $\tan x$ and $\cot x$ are defined and all four are genuine solutions.

Sum of the values of $2x$: $(\pi + \theta) + (2\pi - \theta) + (3\pi + \theta) + (4\pi - \theta) = 10\pi$. Hence the sum of the principal solutions is 5π, independent of λ.

(For $\lambda > 2$ the same method gives 3π, which is option (A).)

Correct Option: (C)

Question 02

English

The number of solutions of the equation $\tan^2 x + \cot^2 x - 3(\tan x - \cot x) = 2$ in the interval $(0, \pi)$ is

(A) 4
(B) 2
(C) 3
(D) 6

Only one option is correct.

हिन्दी

अन्तराल $(0, \pi)$ में समीकरण $\tan^2 x + \cot^2 x - 3(\tan x - \cot x) = 2$ के हलों की संख्या है

(A) 4
(B) 2
(C) 3
(D) 6

केवल एक विकल्प सही है।

Solution

Both $\tan x$ and $\cot x$ must be defined, so $x \neq \frac{\pi}{2}$; in $(0, \pi)$ every other x is allowed.

Let $u = \tan x - \cot x$. Since $\tan x \cot x = 1$, $u^2 = \tan^2 x + \cot^2 x - 2$, so $\tan^2 x + \cot^2 x = u^2 + 2$.

The equation becomes $u^2 + 2 - 3u = 2$, i.e. $u^2 - 3u = 0$, so $u = 0$ or $u = 3$.

Case $u = 0$: $\tan x = \cot x \Rightarrow \tan^2 x = 1 \Rightarrow \tan x = \pm 1$. In $(0, \pi)$: $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$.

Case $u = 3$: with $t = \tan x$, $t - \frac{1}{t} = 3 \Rightarrow t^2 - 3t - 1 = 0$, discriminant $13 > 0$, roots $t = \frac{3 \pm \sqrt{13}}{2}$.

Their product is -1, so one root is positive and one is negative. On $(0, \frac{\pi}{2})$ the tangent takes every positive value exactly once and on $(\frac{\pi}{2}, \pi)$ every negative value exactly once, so each root gives exactly one x.

The two cases give different x (values of u differ), so the number of solutions is $2 + 2 = 4$.

Correct Option: (A)

Question 03

English

Let S be the set of all $x \in [0, 2\pi]$, $x \neq \dfrac{\pi}{2}, \dfrac{3\pi}{2}$, such that $\tan x > 2\sin x$. Then S is a union of intervals whose total length is

(A) $\dfrac{\pi}{2}$

(B) π

(C) $\dfrac{4\pi}{3}$

(D) $\dfrac{2\pi}{3}$

Only one option is correct.

हिन्दी

माना कि S उन सभी $x \in [0, 2\pi]$, $x \neq \dfrac{\pi}{2}, \dfrac{3\pi}{2}$, का समुच्चय है जिनके लिए $\tan x > 2\sin x$ है। तब S अन्तरालों का एक सम्मिलन है जिनकी कुल लम्बाई है

(A) $\dfrac{\pi}{2}$

(B) π

(C) $\dfrac{4\pi}{3}$

(D) $\dfrac{2\pi}{3}$

केवल एक विकल्प सही है।

Solution

For $\cos x \neq 0$: $\tan x - 2\sin x = \dfrac{\sin x - 2\sin x \cos x}{\cos x} = \dfrac{\sin x\,(1 - 2\cos x)}{\cos x}$.

The factors vanish or change sign at: $\sin x = 0$ ($x = 0, \pi, 2\pi$), $\cos x = \frac{1}{2}$ ($x = \frac{\pi}{3}, \frac{5\pi}{3}$), $\cos x = 0$ ($x = \frac{\pi}{2}, \frac{3\pi}{2}$). At $x = 0, \pi, 2\pi$ the difference is 0, so these points are not in S.

$(0, \frac{\pi}{3})$: $\sin x > 0$, $1 - 2\cos x < 0$, $\cos x > 0$, so the expression is negative.

$(\frac{\pi}{3}, \frac{\pi}{2})$: $\sin x > 0$, $1 - 2\cos x > 0$, $\cos x > 0$: positive.

$(\frac{\pi}{2}, \pi)$: $\sin x > 0$, $1 - 2\cos x > 0$, $\cos x < 0$: negative.

$(\pi, \frac{3\pi}{2})$: $\sin x < 0$, $1 - 2\cos x > 0$, $\cos x < 0$: positive.

$(\frac{3\pi}{2}, \frac{5\pi}{3})$: $\sin x < 0$, $1 - 2\cos x > 0$ (as $0 < \cos x < \frac{1}{2}$), $\cos x > 0$: negative.

$(\frac{5\pi}{3}, 2\pi)$: $\sin x < 0$, $1 - 2\cos x < 0$, $\cos x > 0$: positive.

Hence $S = \left(\frac{\pi}{3}, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right)$ with total length $\frac{\pi}{6} + \frac{\pi}{2} + \frac{\pi}{3} = \pi$.

Multiplying both sides by $\cos x$ without regard to its sign would lose the interval $(\pi, \frac{3\pi}{2})$ and give $\frac{\pi}{2}$, option (A).

Correct Option: (B)

Question 04

English

Let S be the set of all $x \in [0, 2\pi]$ for which $\tan x$ and $\cot x$ are defined and which satisfy both the inequalities

$$2\sin^2 x - 3\sin x + 1 < 0 \quad \text{and} \quad \tan x > \cot x.$$

Then S is a union of intervals whose total length is

(A) $\dfrac{\pi}{4}$

(B) $\dfrac{5\pi}{12}$

(C) $\dfrac{\pi}{6}$

(D) $\dfrac{\pi}{3}$

Only one option is correct.

हिन्दी

माना कि S उन सभी $x \in [0, 2\pi]$ का समुच्चय है जिनके लिए $\tan x$ तथा $\cot x$ परिभाषित हैं और जो दोनों असमिकाओं

$$2\sin^2 x - 3\sin x + 1 < 0$$

तथा

$$\tan x > \cot x$$

को सन्तुष्ट करते हैं। तब S अन्तरालों का एक सम्मिलन है जिनकी कुल लम्बाई है

(A) $\dfrac{\pi}{4}$

(B) $\dfrac{5\pi}{12}$

(C) $\dfrac{\pi}{6}$

(D) $\dfrac{\pi}{3}$

केवल एक विकल्प सही है।

Solution

First inequality: $2\sin^2 x - 3\sin x + 1 = (2\sin x - 1)(\sin x - 1) < 0$. Since $\sin x - 1 \le 0$, this needs $\sin x \neq 1$ and $2\sin x - 1 > 0$, i.e. $\frac{1}{2} < \sin x < 1$.

In $[0, 2\pi]$ this is $x \in \left(\frac{\pi}{6}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{5\pi}{6}\right)$.

Second inequality: with $t = \tan x$ ($t \neq 0$), $t > \frac{1}{t} \iff \frac{t^2-1}{t} > 0 \iff (t-1)(t+1)t > 0 \iff t > 1$ or $-1 < t < 0$.

On $\left(\frac{\pi}{6}, \frac{\pi}{2}\right)$ the tangent is positive, so we need $\tan x > 1$, i.e. $x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$.

On $\left(\frac{\pi}{2}, \frac{5\pi}{6}\right)$ the tangent is negative and increases from $-\infty$ to $-\frac{1}{\sqrt{3}}$; we need $-1 < \tan x < 0$, i.e. $x \in \left(\frac{3\pi}{4}, \frac{5\pi}{6}\right)$.

So $S = \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{4}, \frac{5\pi}{6}\right)$, of total length $\frac{\pi}{4} + \frac{\pi}{12} = \frac{\pi}{3}$.

Reading $\tan x > \cot x$ as $\tan^2 x > 1$ (multiplying by $\tan x$ without its sign) keeps only the first interval and gives $\frac{\pi}{4}$, option (A).

Correct Option: (D)

Question 05

English

The number of solutions of the equation $\tan(\pi \sin x) = 1$ in the interval $[0, 9\pi]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[0, 9\pi]$ में समीकरण $\tan(\pi \sin x) = 1$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\theta = \pi \sin x$; then $\theta \in [-\pi, \pi]$. The general solution of $\tan\theta = 1$ is $\theta = n\pi + \frac{\pi}{4}$, $n \in \mathbb{Z}$.

In $[-\pi, \pi]$ this gives $\theta = \frac{\pi}{4}$ $(n = 0)$ and $\theta = -\frac{3\pi}{4}$ $(n = -1)$. Taking only the principal value $\frac{\pi}{4}$ would miss the second family.

So $\sin x = \frac{1}{4}$ or $\sin x = -\frac{3}{4}$. For both, $\theta \neq \pm\frac{\pi}{2}$, so the tangent is defined.

In one period $[2k\pi, 2k\pi + 2\pi)$: $\sin x = \frac{1}{4}$ has two solutions, both in $(2k\pi, 2k\pi + \pi)$; $\sin x = -\frac{3}{4}$ has two solutions, both in $(2k\pi + \pi, 2k\pi + 2\pi)$. That is 4 solutions per period, none at the ends of a period.

$[0, 8\pi]$ contains four full periods: 16 solutions.

On $[8\pi, 9\pi]$, $\sin x \geq 0$, so only $\sin x = \frac{1}{4}$ contributes, with 2 solutions; $\sin x = -\frac{3}{4}$ contributes none.

Total number of solutions $= 16 + 2 = 18$.

Final Answer: 18

Question 06

English

Let S be the set of all $x \in (0, 2\pi)$ satisfying the equation

$$3\sin 2x - 4\cos 2x = 8\sin x - 4\cos x.$$

Then the value of $\sin\left(\sum_{x \in S} x\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी $x \in (0, 2\pi)$ का समुच्चय है जो समीकरण

$$3\sin 2x - 4\cos 2x = 8\sin x - 4\cos x$$

को सन्तुष्ट करते हैं। तब $\sin\left(\sum_{x \in S} x\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Use $\sin 2x = 2\sin x \cos x$ and $\cos 2x = \cos^2 x - \sin^2 x$: the left side minus the right side is $6\sin x \cos x - 4\cos^2 x + 4\sin^2 x - 8\sin x + 4\cos x$.

The quadratic part factors: $4\sin^2 x + 6\sin x \cos x - 4\cos^2 x = 2(2\sin x - \cos x)(\sin x + 2\cos x)$, and the linear part is $-8\sin x + 4\cos x = -4(2\sin x - \cos x)$.

So the equation is $2(2\sin x - \cos x)(\sin x + 2\cos x - 2) = 0$.

Factor 1: $2\sin x = \cos x$. Here $\cos x \neq 0$ (else $\sin x = 0$ too), so $\tan x = \frac{1}{2}$. Let $\alpha = \tan^{-1}\frac{1}{2} \in \left(0, \frac{\pi}{2}\right)$; in $(0, 2\pi)$: $x = \alpha$ and $x = \pi + \alpha$.

Factor 2: $\sin x + 2\cos x = 2$. With $t = \tan\frac{x}{2}$ (the point $x = \pi$ gives $-2 \neq 2$, so nothing is lost): $\frac{2t + 2(1-t^2)}{1+t^2} = 2 \Rightarrow 2t + 2 - 2t^2 = 2 + 2t^2 \Rightarrow 2t(1-2t) = 0$.

$t = 0$ gives $x = 2n\pi$, not in $(0, 2\pi)$. $t = \frac{1}{2}$ gives $\sin x = \frac{2t}{1+t^2} = \frac{4}{5}$, $\cos x = \frac{1-t^2}{1+t^2} = \frac{3}{5}$: one solution $x = \beta \in \left(0, \frac{\pi}{2}\right)$ with $\tan\beta = \frac{4}{3}$.

Since $\tan 2\alpha = \dfrac{2\cdot\frac{1}{2}}{1-\frac{1}{4}} = \frac{4}{3}$ and $2\alpha \in \left(0, \frac{\pi}{2}\right)$, $\beta = 2\alpha$. The three solutions are distinct.

$\sum_{x\in S} x = \alpha + (\pi + \alpha) + 2\alpha = \pi + 4\alpha$.

$\sin(\pi + 4\alpha) = -\sin 4\alpha = -2\sin 2\alpha\cos 2\alpha = -2\cdot\frac{4}{5}\cdot\frac{3}{5} = -\frac{24}{25} = -0.96$.

Final Answer: -0.96

Question 07

English

The sum of all solutions of the equation

$$\sin 3x - \sin x + \sqrt{3}\,(\cos 3x + \cos x) = 2\cos 2x$$

lying in the interval $[-\pi, 0]$ is $\dfrac{k\pi}{6}$. Then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[-\pi, 0]$ में स्थित समीकरण

$$\sin 3x - \sin x + \sqrt{3}\,(\cos 3x + \cos x) = 2\cos 2x$$

के सभी हलों का योग $\dfrac{k\pi}{6}$ है। तब k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Sum-to-product: $\sin 3x - \sin x = 2\cos 2x \sin x$ and $\cos 3x + \cos x = 2\cos 2x\cos x$.

So the left side is $2\cos 2x\left(\sin x + \sqrt{3}\cos x\right)$ and the equation becomes $2\cos 2x\left(\sin x + \sqrt{3}\cos x - 1\right) = 0$.

Factor 1: $\cos 2x = 0 \Rightarrow 2x = \frac{\pi}{2} + n\pi \Rightarrow x = \frac{\pi}{4} + \frac{n\pi}{2}$. In $[-\pi, 0]$: $x = -\frac{\pi}{4}$ $(n = -1)$ and $x = -\frac{3\pi}{4}$ $(n = -2)$.

Factor 2: $\sin x + \sqrt{3}\cos x = 1$. Divide by $\sqrt{1^2 + (\sqrt{3})^2} = 2$: $\frac{1}{2}\sin x + \frac{\sqrt{3}}{2}\cos x = \frac{1}{2}$, i.e. $\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$.

For $x \in [-\pi, 0]$, $x + \frac{\pi}{3} \in \left[-\frac{2\pi}{3}, \frac{\pi}{3}\right]$. In this interval $\sin\theta = \frac{1}{2}$ only at $\theta = \frac{\pi}{6}$ ($\frac{5\pi}{6}$ and $-\frac{7\pi}{6}$ lie outside). So $x = -\frac{\pi}{6}$.

Check: $\cos 2\left(-\frac{\pi}{6}\right) = \frac{1}{2} \neq 0$, so the three solutions are distinct.

Sum $= -\frac{\pi}{4} - \frac{3\pi}{4} - \frac{\pi}{6} = -\pi - \frac{\pi}{6} = -\frac{7\pi}{6}$, so $k = -7$.

Final Answer: -7

Question 08

English

Let S be the set of all $x \in (0, 2\pi)$, $x \neq \pi$, satisfying the equation $4\sin x - 3\cos x - 3 = \tan\dfrac{x}{2}$. Then the value of $\displaystyle\sum_{x\in S} \sin x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी $x \in (0, 2\pi)$, $x \neq \pi$, का समुच्चय है जो समीकरण $4\sin x - 3\cos x - 3 = \tan\dfrac{x}{2}$ को सन्तुष्ट करते हैं। तब $\displaystyle\sum_{x\in S} \sin x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $t = \tan\frac{x}{2}$. As x runs over $(0, 2\pi)$ with $x \neq \pi$, $\frac{x}{2}$ runs over $(0, \pi)$ with $\frac{x}{2} \neq \frac{\pi}{2}$, and t takes every real value exactly once.

Substitute $\sin x = \frac{2t}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$ and multiply by $1 + t^2 > 0$:

$8t - 3(1 - t^2) - 3(1 + t^2) = t(1 + t^2)$, i.e. $8t - 6 = t + t^3$, i.e. $t^3 - 7t + 6 = 0$.

$t = 1$ is a root; dividing, $t^3 - 7t + 6 = (t - 1)(t^2 + t - 6) = (t - 1)(t - 2)(t + 3)$. So $t \in \{1, 2, -3\}$, giving three distinct x in S.

$t = 1$: $\sin x = \frac{2}{2} = 1$ (this is $x = \frac{\pi}{2}$; check: $4 - 0 - 3 = 1 = \tan\frac{\pi}{4}$).

$t = 2$: $\sin x = \frac{4}{5}$, $\cos x = -\frac{3}{5}$; check: $\frac{16}{5} + \frac{9}{5} - 3 = 2$.

$t = -3$: $\sin x = -\frac{6}{10} = -\frac{3}{5}$, $\cos x = -\frac{8}{10} = -\frac{4}{5}$; check: $-\frac{12}{5} + \frac{12}{5} - 3 = -3$.

$$\sum_{x\in S} \sin x = 1 + \frac{4}{5} - \frac{3}{5} = \frac{6}{5} = 1.2.$$

Final Answer: 1.2

Question 09

English

Let x be a real number satisfying $\log_{\cos x}(6\cos x - 1) = \dfrac{1}{2}$. Then the value of $\cos x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि x एक वास्तविक संख्या है जो $\log_{\cos x}(6\cos x - 1) = \dfrac{1}{2}$ को सन्तुष्ट करती है। तब $\cos x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $\log_{\cos x}(6\cos x - 1)$ to be defined we need $\cos x > 0$, $\cos x \neq 1$ and $6\cos x - 1 > 0$.
Under these conditions the equation means $6\cos x - 1 = (\cos x)^{1/2} = \sqrt{\cos x}$.
Let $c = \cos x$. Squaring $6c - 1 = \sqrt{c}$ gives $36c^2 - 12c + 1 = c$, i.e. $36c^2 - 13c + 1 = 0$.
Discriminant $169 - 144 = 25$, so $c = \dfrac{13 \pm 5}{72}$, i.e. $c = \dfrac{1}{4}$ or $c = \dfrac{1}{9}$.
Squaring can add roots of $6c - 1 = -\sqrt{c}$. For $c = \frac{1}{9}$: $6c - 1 = -\frac{1}{3}$, while $\sqrt{c} = \frac{1}{3}$; so it satisfies the squared equation only. It also makes the argument of the logarithm negative. Rejected.
For $c = \frac{1}{4}$: $6c - 1 = \frac{1}{2} = \sqrt{\frac{1}{4}}$; the base $\frac{1}{4}$ is positive and not 1, the argument $\frac{1}{2}$ is positive. Accepted.
Such x exist since $\frac{1}{4} \in (-1, 1)$.
Hence $\cos x = \frac{1}{4} = 0.25$.

Final Answer: 0.25

Question 10

English

The number of solutions of the equation $\sqrt{2-2\sin 3x}\,\sec 3x = 2$ in the interval $[0, 100\pi]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[0, 100\pi]$ में समीकरण $\sqrt{2-2\sin 3x}\,\sec 3x = 2$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\theta = 3x$; $x \in [0, 100\pi]$ corresponds one-to-one to $\theta \in [0, 300\pi]$.

$\sec\theta$ must be defined: $\cos\theta \neq 0$. Then the equation is $\sqrt{2-2\sin\theta} = 2\cos\theta$, whose left side is ≥ 0, so $\cos\theta > 0$.

Squaring (allowed now, both sides being non-negative): $2 - 2\sin\theta = 4\cos^2\theta = 4 - 4\sin^2\theta$, i.e. $2\sin^2\theta - \sin\theta - 1 = 0$, i.e. $(\sin\theta - 1)(2\sin\theta + 1) = 0$.

$\sin\theta = 1$ gives $\cos\theta = 0$: $\sec\theta$ is undefined, so these θ are rejected. (They would satisfy $\sqrt{2-2\sin\theta} = 2\cos\theta$, since both sides are 0.)

$\sin\theta = -\frac{1}{2}$ gives $\theta = \frac{7\pi}{6} + 2n\pi$ or $\theta = \frac{11\pi}{6} + 2n\pi$. At $\frac{7\pi}{6}$, $\cos\theta = -\frac{\sqrt{3}}{2} < 0$: extraneous. At $\frac{11\pi}{6}$, $\cos\theta = \frac{\sqrt{3}}{2}$ and $\sqrt{2+1} = \sqrt{3} = 2\cdot\frac{\sqrt{3}}{2}$: valid.

Count $\theta = \frac{11\pi}{6} + 2n\pi \in [0, 300\pi]$: $0 \leq \frac{11}{6} + 2n \leq 300 \Rightarrow -\frac{11}{12} \leq n \leq 149.08$, so $n = 0, 1, \ldots, 149$.

Hence there are 150 solutions $x = \frac{\theta}{3}$ in $[0, 100\pi]$.

Final Answer: 150

Question 11

English

Consider $f(x) = \sqrt{-\tan x} + \log_e\left(4\pi^2 - x^2\right)$. Let D be the set of all real numbers x for which $f(x)$ is defined, and let $S = \{x \in D : \sin 2x + \sin x = 0\}$. If the sum of all elements of S is $\dfrac{k\pi}{3}$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$f(x) = \sqrt{-\tan x} + \log_e\left(4\pi^2 - x^2\right)$ पर विचार कीजिए। माना कि D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $f(x)$ परिभाषित है, तथा $S = \{x \in D : \sin 2x + \sin x = 0\}$ है। यदि S के सभी अवयवों का योग $\dfrac{k\pi}{3}$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Domain of f: the logarithm needs $4\pi^2 - x^2 > 0$, i.e. $-2\pi < x < 2\pi$. The root needs $\tan x$ to be defined ($x \neq \frac{\pi}{2} + n\pi$) and $-\tan x \geq 0$, i.e. $\tan x \leq 0$.

So $D = \{x \in (-2\pi, 2\pi) : \tan x \leq 0\}$; note that points with $\tan x = 0$ are included.

The equation: $\sin 2x + \sin x = 2\sin x\cos x + \sin x = \sin x\,(2\cos x + 1) = 0$.

Case $\sin x = 0$: in $(-2\pi, 2\pi)$, $x = -\pi, 0, \pi$ (the points $\pm 2\pi$ are outside D because of the logarithm). At each, $\tan x = 0$, so $\sqrt{-\tan x} = 0$ is defined. All three are in S.

Case $\cos x = -\frac{1}{2}$: in $(-2\pi, 2\pi)$, $x = \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{2\pi}{3}, -\frac{4\pi}{3}$.
$\tan\frac{2\pi}{3} = -\sqrt{3} \le 0$: kept. $\tan\frac{4\pi}{3} = \sqrt{3} > 0$: rejected. $\tan\left(-\frac{2\pi}{3}\right) = \sqrt{3} > 0$: rejected. $\tan\left(-\frac{4\pi}{3}\right) = -\sqrt{3} \le 0$: kept.
$S = \left\{-\frac{4\pi}{3}, -\pi, 0, \frac{2\pi}{3}, \pi\right\}$ and its sum is $-\frac{4\pi}{3} + \frac{2\pi}{3} = -\frac{2\pi}{3}$.
Hence $\frac{k\pi}{3} = -\frac{2\pi}{3}$, so $k = -2$.

Final Answer: -2

Question 12

English

The number of solutions of the equation $\cos(\pi x) = \dfrac{x^2}{36}$ in the interval $[-5, 6]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[-5, 6]$ में समीकरण $\cos(\pi x) = \dfrac{x^2}{36}$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Compare the graphs of $y = \cos\pi x$ and $y = \frac{x^2}{36}$. The cosine has period 2, equals 1 at even integers and -1 at odd integers, and is monotonic on each interval $[m, m+1]$, $m \in \mathbb{Z}$.

The parabola is 0 at $x = 0$, increases with $|x|$, and is < 1 for $|x| < 6$; it equals 1 only at $x = \pm 6$.

On $[m, m+1]$ with $-5 \le m \le 4$, the cosine runs monotonically from 1 to -1 or from -1 to 1, while the parabola stays strictly inside $(0, 1)$ and changes by at most $\frac{9}{36}$. So $g(x) = \cos\pi x - \frac{x^2}{36}$ changes sign on each such interval. The cosine is much steeper than the parabola (slope up to π against at most $\frac{1}{3}$), so the graphs cross exactly once there.

Negative side $[-5, 0]$: intervals $[-5, -4], [-4, -3], [-3, -2], [-2, -1], [-1, 0]$, one root each: 5 roots. At $x = 0$: $1 \ne 0$.

Positive side: $(0, 1), (1, 2), (2, 3), (3, 4), (4, 5)$ give 5 roots. On $[5, 6]$: $g(5) = -1 - \frac{25}{36} < 0$, and near $x = 6$, $\cos\pi x \approx 1 - \frac{\pi^2(6-x)^2}{2}$ while $\frac{x^2}{36} \approx 1 - \frac{6-x}{3}$, so $g > 0$ just left of 6: one crossing in $(5, 6)$; and $x = 6$ itself is a root since $\cos 6\pi = 1 = \frac{36}{36}$.

Total $5 + 5 + 2 = 12$.

Final Answer: 12

Question 13

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \max\{\sin|x|,\ \cos|x|\}$. Let S be the set of all $x \in [-2\pi, \pi]$ such that $f(x) = -\dfrac{1}{2}$. If the sum of all elements of S is $k\pi$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \max\{\sin|x|,\ \cos|x|\}$ द्वारा परिभाषित है। माना कि S उन सभी $x \in [-2\pi, \pi]$ का समुच्चय है जिनके लिए $f(x) = -\dfrac{1}{2}$ है। यदि S के सभी अवयवों का योग $k\pi$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since f depends on $|x|$, $f(x) = g(|x|)$ where $g(t) = \max\{\sin t, \cos t\}$; the graph of f for $x < 0$ is the mirror image of that of g for $t > 0$.

On $[0, 2\pi]$, $\sin t \geq \cos t$ exactly on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$. So $g(t) = \cos t$ on $\left[0, \frac{\pi}{4}\right]$, $g(t) = \sin t$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$, and $g(t) = \cos t$ on $\left[\frac{5\pi}{4}, 2\pi\right]$.

The graph of g falls from 1 to $\frac{1}{\sqrt{2}}$, rises to 1 at $\frac{\pi}{2}$, falls to its least value $-\frac{1}{\sqrt{2}}$ at $\frac{5\pi}{4}$, then rises to 1 at 2π. As $-\frac{1}{\sqrt{2}} < -\frac{1}{2}$, the line $y = -\frac{1}{2}$ meets it twice in $[0, 2\pi]$, once on each side of $\frac{5\pi}{4}$.

On the sine branch: $\sin t = -\frac{1}{2}$, $t \in \left[\frac{\pi}{4}, \frac{5\pi}{4}\right] \Rightarrow t = \frac{7\pi}{6}$. On the cosine branch: $\cos t = -\frac{1}{2}$, $t \in \left[\frac{5\pi}{4}, 2\pi\right] \Rightarrow t = \frac{4\pi}{3}$.

The other roots of $\sin t = -\frac{1}{2}$ and $\cos t = -\frac{1}{2}$ ($t = \frac{11\pi}{6}$, $t = \frac{2\pi}{3}$) lie on the wrong branch: there $g = \frac{\sqrt{3}}{2}$.

For $x \in [0, \pi]$, $t = x \in [0, \pi]$ gives $\sin t \geq 0$, so $f(x) \geq 0$: no solutions.

For $x \in [-2\pi, 0)$, $t = -x \in (0, 2\pi]$: $x = -\frac{7\pi}{6}$ and $x = -\frac{4\pi}{3}$.

Sum $= -\frac{7\pi}{6} - \frac{8\pi}{6} = -\frac{15\pi}{6} = -\frac{5\pi}{2}$, so $k = -2.5$.

Final Answer: -2.5

Question 14

English

The sum of all real numbers x satisfying the equation

$$4^{\sin^2 \pi x} + 4^{\cos^2 \pi x} = 4 - |4x - 3|$$

is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

समीकरण

$$4^{\sin^2 \pi x} + 4^{\cos^2 \pi x} = 4 - |4x - 3|$$

को सन्तुष्ट करने वाली सभी वास्तविक संख्याओं x का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Both terms on the left are positive, so by AM $\geq$ GM: $4^{\sin^2 \pi x} + 4^{\cos^2 \pi x} \geq 2\sqrt{4^{\sin^2 \pi x} \cdot 4^{\cos^2 \pi x}} = 2\sqrt{4^1} = 4$.

Equality holds exactly when $4^{\sin^2 \pi x} = 4^{\cos^2 \pi x}$, i.e. $\sin^2 \pi x = \cos^2 \pi x = \frac{1}{2}$, i.e. $x = \frac{1}{4} + \frac{n}{2}$, $n \in \mathbb{Z}$.

The right side satisfies $4 - |4x - 3| \leq 4$, with equality exactly when $x = \frac{3}{4}$.

Since left side $\geq 4 \geq$ right side, the equation holds only if both equal 4: $x = \frac{3}{4}$ and $x \in \left\{\frac{1}{4} + \frac{n}{2}\right\}$.

$x = \frac{3}{4} = \frac{1}{4} + \frac{1}{2}$ is of this form ($\sin^2 \frac{3\pi}{4} = \frac{1}{2}$), so $x = \frac{3}{4}$ is the only solution.

The sum of all solutions is 0.75.

Final Answer: 0.75

Question 15

English

The number of solutions of the equation $\sqrt{3}\sin x + \cos x + \sin\dfrac{3x}{2} = 3$ in the interval $[0, 500\pi]$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अन्तराल $[0, 500\pi]$ में समीकरण $\sqrt{3}\sin x + \cos x + \sin\dfrac{3x}{2} = 3$ के हलों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Range of $a\sin x + b\cos x$: $\sqrt{3}\sin x + \cos x = 2\left(\frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x\right) = 2\sin\left(x + \frac{\pi}{6}\right) \in [-2, 2]$.

Also $\sin\frac{3x}{2} \leq 1$. Hence the left side is at most $2 + 1 = 3$, and the equation holds exactly when both terms take their maximum values at the same x.

$2\sin\left(x + \frac{\pi}{6}\right) = 2 \iff x + \frac{\pi}{6} = \frac{\pi}{2} + 2n\pi \iff x = \frac{\pi}{3} + 2n\pi$, $n \in \mathbb{Z}$.

For these x: $\frac{3x}{2} = \frac{\pi}{2} + 3n\pi$, so $\sin\frac{3x}{2} = \sin\left(\frac{\pi}{2} + 3n\pi\right) = \cos 3n\pi = (-1)^{3n} = (-1)^n$.

This equals 1 exactly when n is even, $n = 2m$. So the solutions are $x = \frac{\pi}{3} + 4m\pi$, $m \in \mathbb{Z}$.

In $[0, 500\pi]$: $0 \leq \frac{1}{3} + 4m \leq 500 \iff -\frac{1}{12} \leq m \leq 124.9$, so $m = 0, 1, \ldots, 124$.

Number of solutions = 125. (Forgetting the parity condition gives 250.)

Final Answer: 125

Question 16

English

Let x be a real number, not an integral multiple of $\dfrac{\pi}{2}$, such that

$$\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} + \frac{4}{5} = 0.$$

Then the value of $\tan^2 x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि x एक वास्तविक संख्या है, जो $\dfrac{\pi}{2}$ का पूर्णांक गुणज नहीं है, जिसके लिए

$$\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} + \frac{4}{5} = 0$$

है। तब $\tan^2 x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since x is not a multiple of $\frac{\pi}{2}$, $\sin x \neq 0$ and $\cos x \neq 0$.

Triple-angle formulas: $\sin 3x = 3\sin x - 4\sin^3 x$, so $\frac{\sin 3x}{\sin x} = 3 - 4\sin^2 x$; $\cos 3x = 4\cos^3 x - 3\cos x$, so $\frac{\cos 3x}{\cos x} = 4\cos^2 x - 3$.

Adding: $\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} = 4\cos^2 x - 4\sin^2 x = 4\cos 2x$.

The equation becomes $4\cos 2x + \frac{4}{5} = 0$, i.e. $\cos 2x = -\frac{1}{5}$. This value lies in $(-1, 1)$, so solutions exist, and for them $\sin x \neq 0, \cos x \neq 0$ (those would need $\cos 2x = \pm 1$).

$$\tan^2 x = \frac{\sin^2 x}{\cos^2 x} = \frac{1 - \cos 2x}{1 + \cos 2x} = \frac{1 + \frac{1}{5}}{1 - \frac{1}{5}} = \frac{6}{4} = 1.5.$$

The value is the same for every solution.

Final Answer: 1.5

Question 17

English

Let S be the set of all $x \in (-2\pi, 2\pi)$ for which $\sec x$, $\tan x$ and $\tan\dfrac{x}{2}$ are defined and

$$\sec x + \tan x = 1 + 4\tan\frac{x}{2}.$$

Then the value of $\displaystyle\sum_{x \in S} (\sin x + \cos x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी $x \in (-2\pi, 2\pi)$ का समुच्चय है जिनके लिए $\sec x$, $\tan x$ तथा $\tan\dfrac{x}{2}$ परिभाषित हैं और

$$\sec x + \tan x = 1 + 4\tan\frac{x}{2}$$

है। तब $\displaystyle\sum_{x \in S} (\sin x + \cos x)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $t = \tan\frac{x}{2}$ (defined since $x \neq \pm\pi$). Then $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, and $\cos x \neq 0$ means $t \neq \pm 1$.

$$\sec x + \tan x = \frac{1+\sin x}{\cos x} = \frac{(1+t^2)+2t}{1-t^2} = \frac{(1+t)^2}{(1-t)(1+t)} = \frac{1+t}{1-t}.$$

The equation is $\dfrac{1+t}{1-t} = 1+4t$, i.e. $1+t = (1+4t)(1-t) = 1+3t-4t^2$, i.e. $4t^2-2t=0$, so $t=0$ or $t=\frac{1}{2}$ (neither is ± 1).

As x runs over $(-2\pi, 2\pi)$, $\frac{x}{2}$ runs over $(-\pi,\pi)$, an interval of length 2π, on which $\tan\frac{x}{2}$ takes every real value twice except where $\frac{x}{2} = \pm\frac{\pi}{2}$.

$t=0$: $\frac{x}{2} \in \{0\}$ only ($\pm\pi$ are excluded by the open interval), so $x=0$, with $\sin x + \cos x = 0+1 = 1$.

$t=\frac{1}{2}$: $\frac{x}{2} = \alpha$ or $\alpha - \pi$, where $\alpha = \tan^{-1}\frac{1}{2}$; so $x = 2\alpha$ or $2\alpha - 2\pi$. For both, $\sin x = \frac{2t}{1+t^2} = \frac{4}{5}$, $\cos x = \frac{1-t^2}{1+t^2} = \frac{3}{5}$, so $\sin x + \cos x = \frac{7}{5}$. Check: $\sec x + \tan x = \frac{5}{3} + \frac{4}{3} = 3 = 1 + 4\cdot\frac{1}{2}$.

$S = \{2\alpha - 2\pi,\ 0,\ 2\alpha\}$ and $\displaystyle\sum_{x\in S}(\sin x + \cos x) = 1 + \frac{7}{5} + \frac{7}{5} = \frac{19}{5} = 3.8.$

Final Answer: 3.8

Question 18

English

Let x and y be real numbers such that

$$x - y = \frac{\pi}{2} \qquad \text{and} \qquad 3\sin x - 4\sin y = 5.$$

Then the value of $\sin(x+y)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि x तथा y वास्तविक संख्याएँ हैं जिनके लिए

$$x - y = \frac{\pi}{2}$$

तथा

$$3\sin x - 4\sin y = 5$$

हैं। तब $\sin(x+y)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

From the first equation, $y = x - \frac{\pi}{2}$, so $\sin y = \sin\left(x - \frac{\pi}{2}\right) = -\cos x$.

The second equation becomes $3\sin x + 4\cos x = 5$.

Write $3\sin x + 4\cos x = 5\left(\frac{3}{5}\sin x + \frac{4}{5}\cos x\right) = 5\cos(x-\beta)$ with $\cos\beta = \frac{4}{5}$, $\sin\beta = \frac{3}{5}$. The maximum is 5, so the equation needs $\cos(x-\beta) = 1$, i.e. $x = \beta + 2n\pi$.

Hence $\sin x = \frac{3}{5}$ and $\cos x = \frac{4}{5}$ for every solution.

$x + y = 2x - \frac{\pi}{2}$, so $\sin(x+y) = \sin\left(2x - \frac{\pi}{2}\right) = -\cos 2x = -(\cos^2 x - \sin^2 x) = -\left(\frac{16}{25} - \frac{9}{25}\right) = -\frac{7}{25}$.

So $\sin(x+y) = -0.28$.

Final Answer: -0.28